

Homework 3: Distribution shift, learnability, and interpolation

Due Friday, October 16, 2026, at 11:59 p.m. This assignment is worth 100 points and 10% of the course grade. Late homework is not accepted; the two-lowest-grade replacement described in the syllabus covers ordinary illness, overload, and missed work. Complete every exercise or part assigned points. Exercises or parts not assigned points are optional and ungraded. Submit one ZIP archive named hw03-ccid.zip, containing a single top-level directory named hw03-ccid with your PDF, LaTeX source, and any supporting files. Replace ccid with your University of Alberta CCID.

General instructions. Justify every answer, including yes/no answers, even when the question does not explicitly ask for a justification.

Exercise profiles. Filled mountains show difficulty on a four-mountain scale: one is a direct application, two uses a familiar method, three combines ideas, and four is a substantial challenge. Filled cogs show technical load on a three-cog scale: one is light, two is sustained, and three is dense. Filled hourglasses show estimated time: one is under 20 minutes, two is 20–45 minutes, three is 45–90 minutes, and four is more than 90 minutes. Pale pictograms show the unused part of each scale. The work row uses the canonical order  proof,  calculation,  construction,  computation, and  interpretation; pale icons mark work types not used.  marks the intended learning outcome. Time estimates cover the non-optional parts. Ratings are not prerequisites or grading weights. A superscript asterisk marks an exercise based on supplementary lecture material; it is not a difficulty rating and does not make the exercise optional when assigned.

Name: YOUR NAME CCID: YOUR CCID

Points: (a): 13, (b): 11; total 24.

Exercise 1: The cost of a compromise under a changed mixture.

 Quantify how a restricted model's population optimum changes with the training mixture.

For $\lambda \in [0, 1]$, define the input probability mass function μ_λ by

$$\mu_\lambda(a) = \lambda, \quad \mu_\lambda(b) = 1 - \lambda.$$

Let $k(\cdot | x)$ be the conditional probability mass function on $\{0, 1\}$ specified by

$$k(1 | a) = 0.1, \quad k(1 | b) = 0.9, \quad k(0 | x) = 1 - k(1 | x).$$

Define the joint law P_λ on $\{a, b\} \times \{0, 1\}$ through its mass function

$$P_\lambda(x, y) = \mu_\lambda(x)k(y | x).$$

For each $\lambda \in [0, 1]$, let (X, Y) have joint law P_λ . For $q \in (0, 1)$, let $f_q : \{a, b\} \rightarrow (0, 1)$ satisfy $f_q(a) = f_q(b) = q$. Define the constant and context-dependent model classes

$$\mathcal{F}_{\text{const}} = \{f_q : q \in (0, 1)\}, \quad \mathcal{F}_{\text{ctx}} = \{f : \{a, b\} \rightarrow (0, 1)\}.$$

For $f \in \mathcal{F}_{\text{ctx}}$, define

$$L_\lambda^{\text{sq}}(f) = \mathbb{E}[(f(X) - Y)^2], \quad L_\lambda^{\text{log}}(f) = \mathbb{E}[-Y \log f(X) - (1 - Y) \log(1 - f(X))].$$

- (a) Let $q_\lambda^* = 0.9 - 0.8\lambda$. Prove that $f_{q_\lambda^*}$ is the unique minimizer of both losses over $\mathcal{F}_{\text{const}}$. Establish, for every $q \in (0, 1)$,

$$L_\lambda^{\text{sq}}(f_q) - L_\lambda^{\text{sq}}(f_{q_\lambda^*}) = (q - q_\lambda^*)^2, \quad L_\lambda^{\text{log}}(f_q) - L_\lambda^{\text{log}}(f_{q_\lambda^*}) = D(\text{Ber}(q_\lambda^*), \text{Ber}(q)).$$

- (b) Suppose training uses λ and deployment uses $\nu \in [0, 1]$. Derive the deployment excess risks of the training minimizer:

$$L_\nu^{\text{sq}}(f_{q_\lambda}^*) - L_\nu^{\text{sq}}(f_{q_\nu}^*) = 0.64(\lambda - \nu)^2,$$

$$L_\nu^{\text{log}}(f_{q_\lambda}^*) - L_\nu^{\text{log}}(f_{q_\nu}^*) = D(\text{Ber}(q_\nu^*), \text{Ber}(q_\lambda^*)).$$

Define $f^* \in \mathcal{F}_{\text{ctx}}$ by $f^*(a) = 0.1$ and $f^*(b) = 0.9$. Prove that, for every $\rho \in [0, 1]$ and $f \in \mathcal{F}_{\text{ctx}}$,

$$L_\rho^{\text{sq}}(f) - L_\rho^{\text{sq}}(f^*) = \rho(f(a) - 0.1)^2 + (1 - \rho)(f(b) - 0.9)^2$$

and

$$L_\rho^{\text{log}}(f) - L_\rho^{\text{log}}(f^*) = \rho D(\text{Ber}(0.1), \text{Ber}(f(a))) + (1 - \rho) D(\text{Ber}(0.9), \text{Ber}(f(b))).$$

Use these identities to prove that f^* minimizes both risks over $\mathcal{F}_{\text{ctx}}$ for every ρ . Identify the restriction to $\mathcal{F}_{\text{const}}$ as the source of the nonzero deployment excess risks.

HINTS For (a), first determine $\mathbb{P}(Y = 1)$. For the $\mathcal{F}_{\text{ctx}}$ identities, consider the risk at each context separately.

Solution

Write your solution here.

Points: (a): 10, (b): 3, (c): 7, (d): 5; total 25.

Exercise 2: From small gradients to close trajectory laws.

⊙ Combine Gaussian KL, Pinsker's inequality, and a coupling to compare two stochastic optimization trajectories.

Let $F : \mathbb{R}^p \rightarrow \mathbb{R}$ be differentiable, and fix $T \in \mathbb{N}$, $\eta, \sigma, \varepsilon > 0$. Let the $\mathbb{R}^p$ -valued random vectors $\Theta_0, \Xi_0, \dots, \Xi_{T-1}$ be mutually independent, with

$$\mathcal{L}(\Theta_0) = \mathcal{N}(0, \sigma^2 I_p), \quad \mathcal{L}(\Xi_t) = \mathcal{N}(0, \sigma^2 I_p).$$

Define

$$[g]_\varepsilon = \begin{cases} 0, & \|g\| \leq \varepsilon, \\ g, & \|g\| > \varepsilon, \end{cases}$$

and, using the same initialization and increments, define three processes by

$$\begin{aligned} \Theta_{t+1} &= \Theta_t - \eta \nabla F(\Theta_t) - \Xi_t, \\ \bar{\Theta}_0 &= \Theta_0, \quad \bar{\Theta}_{t+1} = \bar{\Theta}_t - \eta [\nabla F(\bar{\Theta}_t)]_\varepsilon - \Xi_t, \\ \Theta_0^{(0)} &= \Theta_0, \quad \Theta_{t+1}^{(0)} = \Theta_t^{(0)} - \Xi_t, \quad 0 \leq t < T. \end{aligned}$$

For any process $V_0, \dots, V_T$, write $V_{0:T} = (V_0, \dots, V_T)$. Let

$$\mathbf{P} = \mathcal{L}(\Theta_{0:T}), \quad \mathbf{R} = \mathcal{L}(\bar{\Theta}_{0:T}), \quad \mathbf{G} = \mathcal{L}(\Theta_{0:T}^{(0)}).$$

Suppose

$$\mathbb{P} \left(\max_{0 \leq t \leq T} \|\nabla F(\Theta_t^{(0)})\| > \varepsilon \right) \leq \delta.$$

- (a) For a fixed common history $\vartheta_{0:t}$, compute the KL divergence between the next-step Gaussian kernels of $\mathbf{P}$ and $\mathbf{R}$. Use the KL chain rule to prove

$$D(\mathbf{P}, \mathbf{R}) \leq \frac{T\eta^2\varepsilon^2}{2\sigma^2}.$$

- (b) Apply Pinsker's inequality to prove

$$\text{TV}(\mathbf{P}, \mathbf{R}) \leq \frac{\eta\varepsilon\sqrt{T}}{2\sigma}.$$

- (c) Couple the auxiliary process and the Gaussian walk by using the same initial condition and Gaussian increments. Prove $\text{TV}(\mathbf{R}, \mathbf{G}) \leq \delta$.
- (d) Combine the two bounds to obtain a bound on $\text{TV}(\mathbf{P}, \mathbf{G})$. Then, for positive d, A, k , substitute $\varepsilon = e^{-k/18}$, $\delta = (T+1)e^{-k/18}$, and

$$\frac{\eta\sqrt{T}}{2\sigma} + T + 1 \leq 3d^{5A/2}, \quad 3d^{5A/2} \leq e^{k/36},$$

to prove

$$\text{TV}(\mathbf{P}, \mathbf{G}) \leq e^{-k/36}.$$

HINTS For (a), Gaussians with common covariance $\sigma^2 I$ and means μ, ν have KL divergence $\|\mu - \nu\|^2 / (2\sigma^2)$. For (c), use the defining rule for the clipped gradient and argue by induction on time.

Solution

Write your solution here.

Points: (a): 10, (b): 11, (c): 6; total 27.

Exercise 3: Learn a two-bit rule and compose it.

⦿ Prove convergence for a locally supervised gate and transfer its correctness to a multistep computation.

For $a, b \in \{0, 1\}$, fix

$$\phi(a, b) = \begin{pmatrix} [a+b]_+ \\ [a+b-1]_+ \end{pmatrix}, \quad g_u(a, b) = u^\top \phi(a, b), \quad u \in \mathbb{R}^2.$$

Only the two output weights u are trained. With equal weight on the four input pairs, let

$$L(u) = \frac{1}{4} \sum_{a, b \in \{0, 1\}} (g_u(a, b) - (a \oplus b))^2.$$

- (a) Prove that the unique minimizer is $u^* = (1, -2)^\top$ and that $L(u^*) = 0$. Derive

$$\nabla^2 L(u) = H = \begin{pmatrix} 3 & 1 \\ 1 & 1/2 \end{pmatrix}, \quad \lambda_{\min}(H) = \frac{7 - \sqrt{41}}{4}, \quad \lambda_{\max}(H) = \frac{7 + \sqrt{41}}{4}.$$

- (b) For any initial $u_0 \in \mathbb{R}^2$, run $u_{t+1} = u_t - \eta \nabla L(u_t)$ with $\eta = 1/\lambda_{\max}(H)$. Prove

$$\|u_t - u^*\| \leq r^t \|u_0 - u^*\|, \quad r = 1 - \frac{\lambda_{\min}(H)}{\lambda_{\max}(H)} \in (0, 1).$$

Let $\hat{g}_t(a, b) = \mathbb{I}\{g_{u_t}(a, b) \geq 1/2\}$. Show that $\hat{g}_t(a, b) = a \oplus b$ on all four inputs whenever $\sqrt{5} r^t \|u_0 - u^*\| < 1/2$.

- (c) For $b_{1:d} \in \{0, 1\}^d$, define $\hat{z}_1 = b_1$ and $\hat{z}_i = \hat{g}_t(\hat{z}_{i-1}, b_i)$ for $2 \leq i \leq d$. Prove that the condition in (b) gives $\hat{z}_d = b_1 \oplus \dots \oplus b_d$ for every input. State separately which components are fixed by the construction and which are learned. Your answer must account for the two ReLU features ϕ , the threshold at $1/2$, the recurrent connections that feed $(\hat{z}_{i-1}, b_i)$ to the next gate, and the output-weight vector u .

HINTS For (a), evaluate ϕ on the four input pairs. For (b), write the recurrence for $u_t - u^*$ and use an orthonormal eigenbasis of the symmetric matrix H . For (c), use induction over the output positions.

Solution

Write your solution here.

Points: (a): 6, (b): 8; total 14.

Exercise 4: Point corrections and the test measure.

- ⊙ Separate exact fitting at the observations from integrated prediction error.

Let μ be a Borel probability law on $\mathcal{X} \subset \mathbb{R}$, let $x_1, \dots, x_n \in \mathcal{X}$ be distinct, and let g, f^* be measurable, square-integrable functions with specified point values. For numbers $r_1, \dots, r_n$, define

$$h(x) = g(x) + \sum_{i=1}^n r_i \mathbb{I}\{x = x_i\}.$$

- (a) Suppose that $\mu(\{x\}) = 0$ for every $x \in \mathcal{X}$. Prove

$$\|h - g\|_{L^2(\mu)} = 0, \quad \|h - f^*\|_{L^2(\mu)} = \|g - f^*\|_{L^2(\mu)}.$$

Choose r_i so that $h(x_i) = y_i$ for arbitrary prescribed values $y_1, \dots, y_n$.

- (b) Show that the conclusion can fail when the test law has a point mass. Give an explicit choice of $\mathcal{X}, \mu, g, f^*, x_1, r_1$ for which

$$\|h - f^*\|_{L^2(\mu)}^2 > \|g - f^*\|_{L^2(\mu)}^2.$$

Show that the distinction from part (a) is whether the set on which the correction is nonzero has zero or positive μ -measure, and state why this determines whether the correction contributes to the population $L^2(\mu)$ error.

- (c) **Optional.** Let $P \in \mathcal{P}_2$ be arbitrary, and assume that every singleton in $\mathcal{X}$ is measurable. Let $S_n = ((X_i, Y_i))_{i=1}^n$ be an i.i.d. sample with $\mathcal{L}(X_i, Y_i) = P$. Write $N_n(x) = \sum_{i=1}^n \mathbb{I}\{X_i = x\}$ and define

$$\hat{f}_n(x) = \begin{cases} N_n(x)^{-1} \sum_{i: X_i=x} Y_i, & N_n(x) > 0, \\ \tilde{f}_n(x), & N_n(x) = 0, \end{cases}$$

where $\tilde{f}_n$ is any estimator. Prove that, for every training index i such that $X_j = X_i$ implies $Y_j = Y_i$ for all $j \in [n]$,

$$\hat{f}_n(X_i) = Y_i.$$

(d) **Optional.** In the setting of part (c), suppose that

$$\|\tilde{f}_n - f_P^*\|_{L^2(P_X)}^2 \xrightarrow{\mathbb{P}} 0.$$

Prove

$$\|\hat{f}_n - f_P^*\|_{L^2(P_X)}^2 \xrightarrow{\mathbb{P}} 0 \quad \text{as } n \rightarrow \infty.$$

You may use that the set $\mathcal{D} = \{x : P_X(\{x\}) > 0\}$ is countable, the strong law of large numbers, and dominated convergence for sums over $\mathcal{D}$.

HINTS For (b), a one-point space is sufficient.

Solution

Write your solution here.

Points: 10.

Exercise 5: Feature scaling can change the minimum-norm interpolant.

🎯 Construct a two-parameter example in which feature scaling changes the minimum-norm interpolant.

Using two parameters, construct two feature maps that represent the same function class but whose minimum-Euclidean-parameter-norm interpolants are different under the same interpolation constraint.

Solution

Write your solution here.